

ECON 5345 Homework 2 Report

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Question 1

a. Taking logarithm, we have

$$\log A_t = z_t + \alpha \log \Gamma_t.$$

Taking difference, we get

$$\Delta a_t = \log A_t - \log A_{t-1} = (\rho_z - 1)z_{t-1} + \alpha g_t + e_t^z.$$

Since $\{z_t\}$ and $\{g_t\}$ are stationary and $e_t^z \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_z^2)$, we have Δa_t being stationary. So $d = 1$.

Note that

$$(1 - \rho_z L)\Delta z_t = (1 - L)e_t^z, \text{ and } (1 - \rho_g L)g_t = (1 - \rho_g)\mu_g + e_t^g.$$

Multiplying $\Delta a_t = \Delta z_t + \alpha \Delta g_t$ by $(1 - \rho_z L)(1 - \rho_g L)$, we get

$$(1 - \rho_z L)(1 - \rho_g L)\Delta a_t = (1 - \rho_g L)(1 - L)e_t^z + \alpha(1 - \rho_z L)[(1 - \rho_g)\mu_g + e_t^g]. \quad (1)$$

The LHS characterizes an $AR(2)$ process and the RHS characterizes a $MA(2)$ process. So $p = 2, q = 2$.

b. Consider the stationary z part first. We have

$$z_{t+h} = \sum_{j=0}^{+\infty} \rho_z^j e_{t+h-j}^z.$$

Given $\{z_\tau\}_{\tau=-\infty}^{t-1}$, the forecast error variance is

$$FEV_h^z = \text{Var} \left[\sum_{j=0}^h \rho_z^j e_{t+h-j}^z \right] = \sigma_z^2 \frac{1 - \rho_z^{2(h+1)}}{1 - \rho_z^2}.$$

Now consider the trend component $\log \Gamma_t$. Note that

$$\log \Gamma_{t+h} = \log \Gamma_{t-1} + \sum_{j=0}^h g_{t+j}.$$

Ignoring the constant term, the g_t process can be written as

$$g_{t+h} = \sum_{j=0}^{+\infty} \rho_g^j e_{t+h-j}^g.$$

Given $\{z_\tau\}_{\tau=-\infty}^{t-1}$, the forecast error variance is

$$FEV_h^\Gamma = \text{Var} \left[\sum_{j=0}^h \sum_{i=0}^j \rho_g^i e_{t+j-i}^g \right].$$

We can compute the values by plugging in the parameter values. Figure 1 shows that plot.

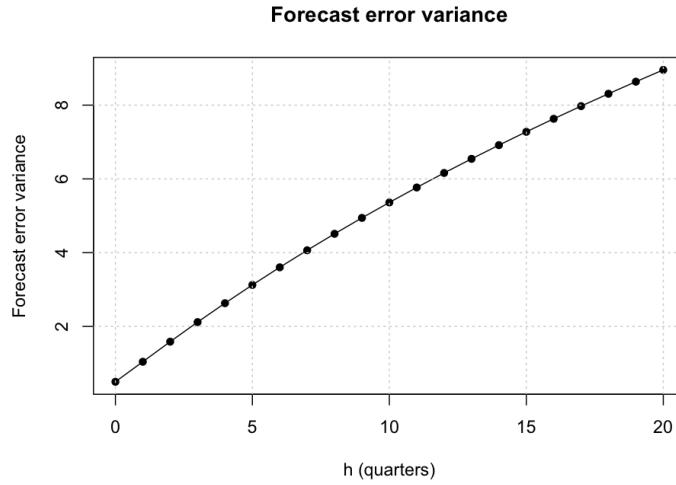


Figure 1: Forecast Error Variance

c. The decomposition is shown in Figure 2.

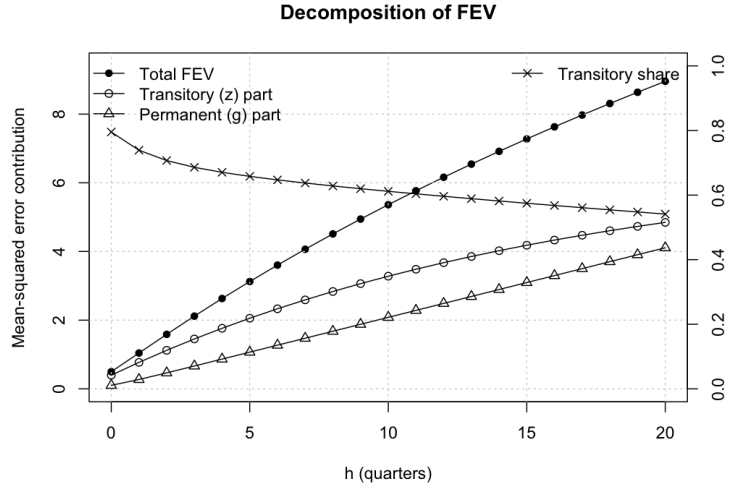


Figure 2: Decomposition of Forecast Error Variance

The effect of transitory components decays with time. The permanent components have a more persistent effect. From the expressions of FEV^z and FEV^Γ , mathematically, we can observe that each term of e^z is summed once, while each term of e^g is summed multiple times. This makes the effect from permanent components accumulate.

Question 2

a. Rewriting the $ARMA(1, 1)$ process, we get

$$(1 - 0.8L)y_t = (1 + 0.4L)e_t.$$

Suppose that $y_t = \psi(L)e_t$. Then we have

$$(1 - 0.8L)(\psi_0 + \psi_1L + \cdots) = 1 + 0.4L.$$

Equating coefficients, we get the impulse responses:

$$\psi_j = \begin{cases} 1, & j = 0 \\ 1.2 \times 0.8^{j-1}, & j \geq 1 \end{cases}$$

Plotting this function, we get Figure 3. Codes in “hw2-q2.R”.

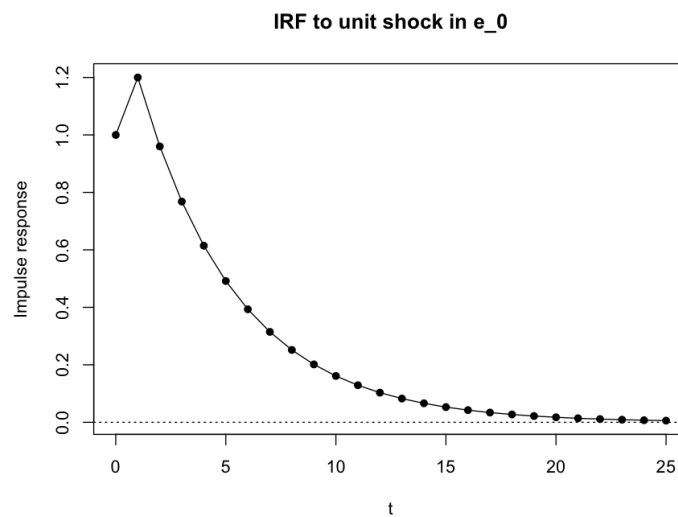


Figure 3: IRF to unit shock in e_0

b. We can write y_t as a $MA(\infty)$ process:

$$y_t = \sum_{j=0}^{\infty} \psi_j e_{t-j}.$$

It immediately follows that $\mathbb{E}[y_t] = 0$.

The variance is

$$\text{Var}[y_t] = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j^2 \right) = 3 \left(1 + \frac{1.2^2}{1 - 0.64} \right) = 15.$$

The autocovariances are

$$\gamma(1) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+1} \right) = 3 \left(1.2 + \frac{1.2^2 \times 0.8}{1 - 0.64} \right) = 13.2.$$

$$\gamma(2) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+2} \right) = 3 \left(0.96 + \frac{1.2^2 \times 0.8^2}{1 - 0.64} \right) = 10.56.$$

c. Since $\text{Var}[e_t] = 3$, we have the ACGF $G : \mathbb{C} \rightarrow \mathbb{C}$ defined as

$$\begin{aligned} G(z) &= 3 \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^j \right) \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^{-j} \right) \\ &= 3 \times \left[\left(1 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2(j-1)} \right) + \left(1.2 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j-1} \right) z \right. \\ &\quad \left. + \left(0.96 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j} \right) z^2 + \dots \right]. \end{aligned}$$

By the ACGF,

$$\gamma(0) = 15, \gamma(1) = 13.2, \gamma(2) = 10.56.$$

This verifies the results in part (b).

- d. I used $e \sim \mathcal{N}(0, 3)$ to generate the innovations. The figure 4 is the simulated path of y . Codes in “hw2_q2.R”.

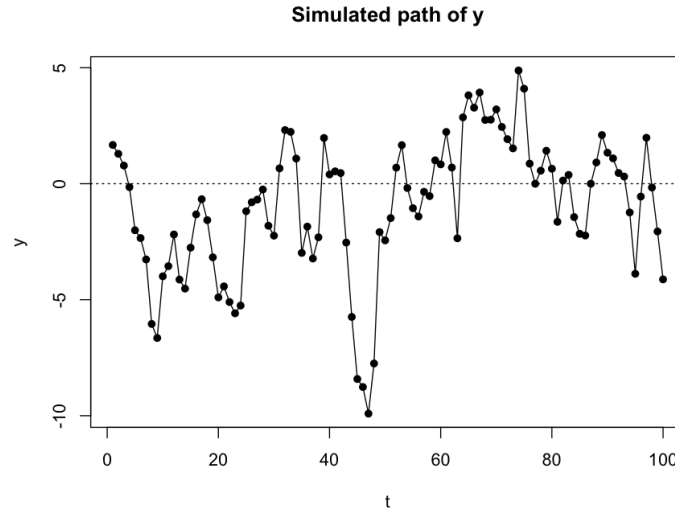


Figure 4: Simulated path of y

The reason for burning is to gain stationarity in the long run.

- e. i. I tried $M = 3$ in the algorithm. The following tables are the reports:

	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	-0.0154	0.0093	-1.65	0.1024
y_lag	0.6891	0.0035	196.14	0.0000
e_lag	0.3882	0.0063	61.57	0.0000

Table 1: Hannan–Rissanen Estimates

- ii. The MoM estimates are $\hat{\alpha} = 0.6785974$ and $\hat{\theta} = 0.4163043$.

f. Table 2 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.76	-0.04	0.01	0.08
HR	theta	0.40	0.40	-0.00	0.01	0.11
MoM	alpha	0.80	0.78	-0.02	0.01	0.08
MoM	theta	0.40	0.44	0.04	0.04	0.19

Table 2: Parameter estimates statistics.

Distributions are in Figure 5.

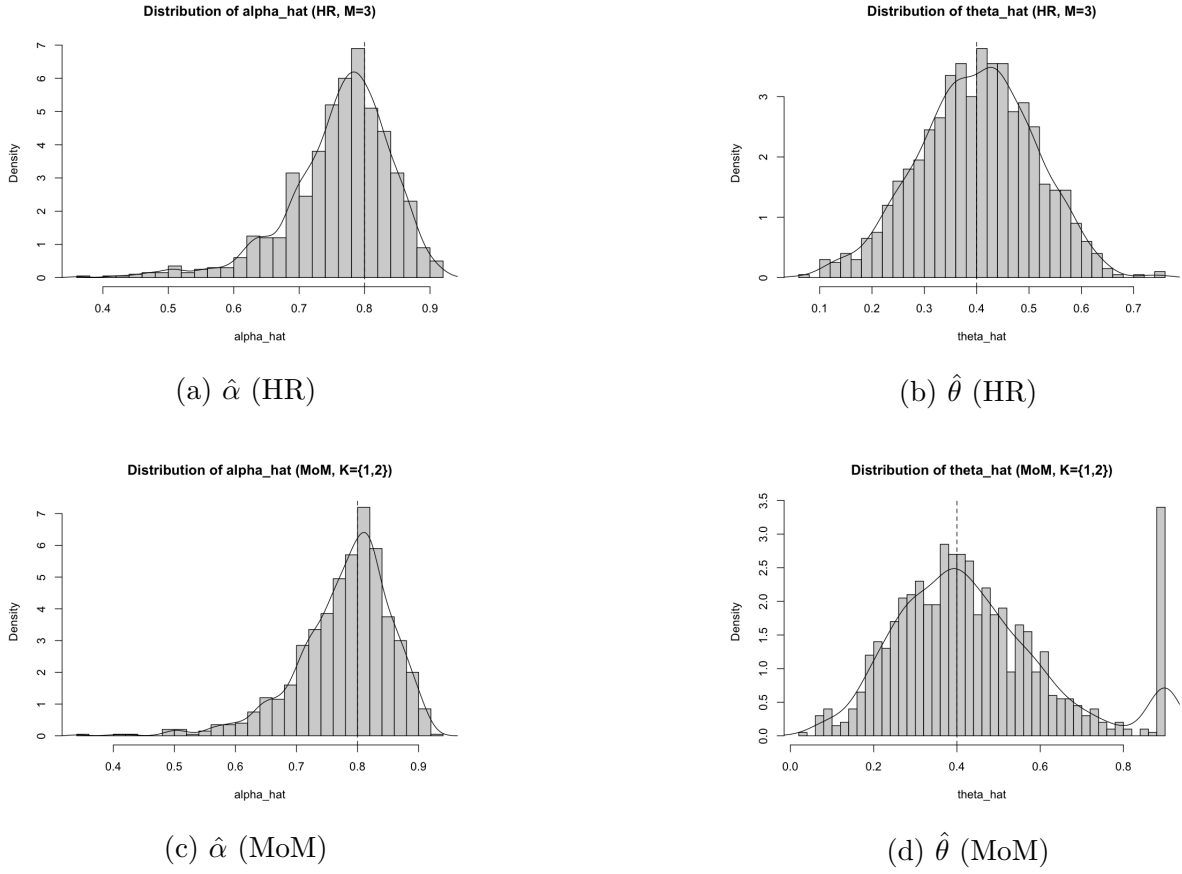


Figure 5: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

g. First, $T = 50$.

Table 3 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.71	-0.09	0.02	0.13
HR	theta	0.40	0.41	0.01	0.02	0.15
MoM	alpha	0.80	0.75	-0.05	0.02	0.12
MoM	theta	0.40	0.46	0.06	0.06	0.24

Table 3: Parameter estimates statistics.

Distributions are in Figure 6.

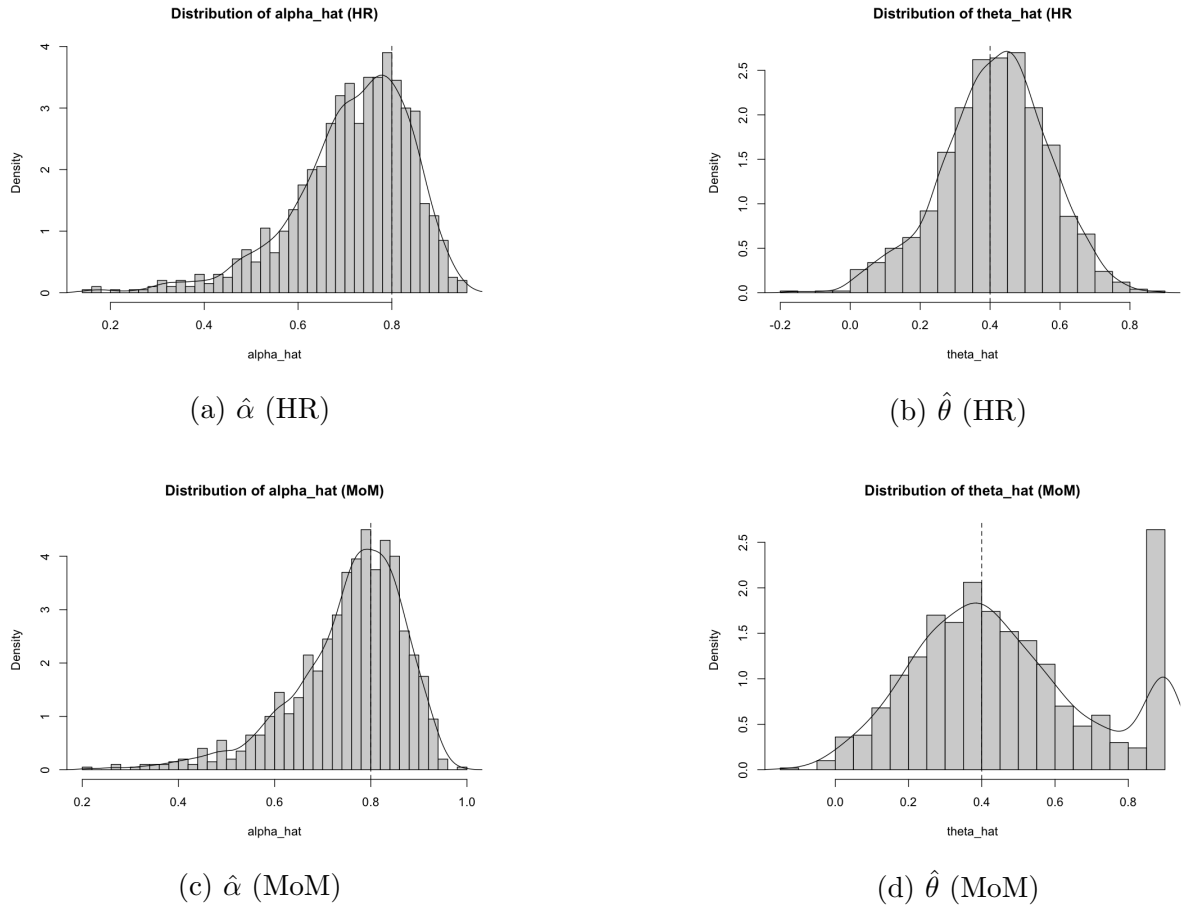


Figure 6: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

Then $T = 500$.

Table 4 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.79	-0.01	0.00	0.03
HR	theta	0.40	0.40	-0.00	0.00	0.05
MoM	alpha	0.80	0.79	-0.01	0.00	0.03
MoM	theta	0.40	0.41	0.01	0.00	0.07

Table 4: Parameter estimates statistics.

Distributions are in Figure 7.

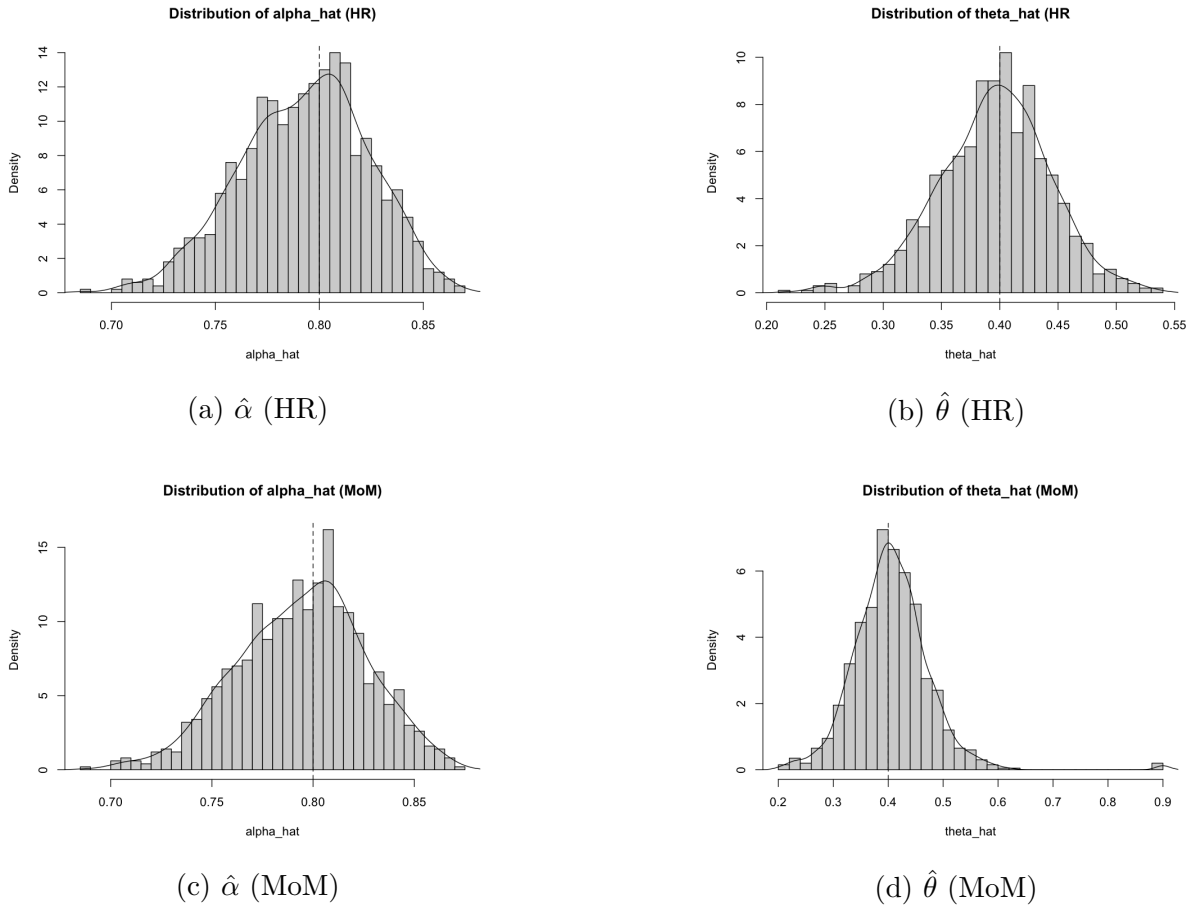


Figure 7: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

- h. As T increases, the mean value of the estimates converges to the true parameter value, the MSE vanishes, and the standard deviation decreases. The distribution are more concentrated around the true parameter value.

I also observe a spike around 1 in the distribution of $\hat{\theta}$ using MoM when T is small. The density at (around) 1 shrinks as T increases. This is because the sample is noisy when the size is small so that the MSE function is flatter and the optimizer will fall around the boundary. The identification of θ is strengthened when the sample size becomes larger.

Question 3

- a. The following figure shows the result.

Question 4

a. The expectation is

$$\begin{aligned}\mathbb{E}[x_t] &= \mathbb{E}[\alpha] \cos t + \mathbb{E}[\beta] \sin t \\ &= 0.\end{aligned}$$

The variance is

$$\begin{aligned}\text{Var}[x_t] &= \text{Var}[\alpha] \cos^2 t + \text{Var}[\beta] \sin^2 t + 2 \text{Cov}(\alpha, \beta) \cos t \sin t \\ &= \cos^2 t + \sin^2 t \\ &= 1.\end{aligned}$$

The second line is by the independence of α and β .

b. No. Note that

$$\begin{aligned}\mathbb{E}[\alpha^2] &= \text{Var}[\alpha] + (\mathbb{E}[\alpha])^2 \\ &= 1 \\ \mathbb{E}[\beta^2] &= \text{Var}[\beta] + (\mathbb{E}[\beta])^2 \\ &= 1 \\ \mathbb{E}[\alpha\beta] &= \text{Cov}(\alpha, \beta) + \mathbb{E}[\alpha]\mathbb{E}[\beta] \\ &= 0.\end{aligned}$$

As a result, we have

$$\begin{aligned}\mathbb{E}[x_t x_{t-k}] &= \mathbb{E}[\alpha^2] \cos t \cos(t-k) + \mathbb{E}[\beta^2] \sin t \sin(t-k) \\ &\quad + \mathbb{E}[\alpha\beta][\cos t \sin(t-k) + \sin t \cos(t-k)] \\ &= \cos t \cos(t-k) + \sin t \sin(t-k) \\ &= \cos[t - (t-k)] \\ &= \cos k.\end{aligned}$$